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OBJECTIVE IDEA ANALYSIS

Invention Name: Retinal Prosthesis and Method of Manufacturing a Retinal Prosthesis

Submitted By: User-provided patent PDF

Invention ID: US 8,527,057

Date of Report: 2026-08-15

This report was generated by the inventionevaluator report-automation platform from the Invention Evaluation Engine v1.6 pipeline (evidence sufficiency architecture). It compiles established findings, analytical conclusions, and an operational audit produced by the nine-phase evaluation. Scores shown are derived executive summaries of the underlying findings — they are presentation-layer derivations, not new evidence, and do not constitute an overall patentability label. This document is not legal advice.

This is a preliminary evidence-constrained evaluation based on a public patent record and selected public sources. It is not legal advice, not a patentability opinion, and not an FTO opinion.

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Executive Summary

Derived executive summary — constrained to Established Findings + Analytical Conclusions (Executive Summary $\subseteq$ findings + conclusions). Gauge and bar scores are presentation-layer derivations from the findings; basis proposition_ids are listed in the scores manifest.

Very Limited Limited Limited-Moderate Moderate Moderate-Strong Strong



Commercialization Summary

placeholder

Evidence trace

The proposition register below is extracted from the compiled report so every score basis can be checked against visible report content.

Technology Analysis

Technology Profile — US 8,527,057 B2

Plain-language description

This is an implantable epiretinal prosthesis that combines a retinal electrode array with a hermetic ASIC package and an inductive receiving coil mounted on a scleral strap. The package architecture uses ceramic vias, flip-chip and wire-bond interconnects, a brazed case wall, vacuum bake-out, and a laser-welded lid to create a compact implant intended to reduce tissue irritation and obstruction of eye motion.

Feature-to-benefit map

Innovation assessment

- The claimed delta is not electrical retinal stimulation by itself; it is the integrated low-profile mechanical arrangement and hermetic package architecture.
- The invention combines known elements from retinal electrodes, inductive coupling, ceramic feedthroughs, flip-chip packaging, and implant fixation. This creates **combination-obviousness exposure**; the relevant question is whether the specific ocular packaging arrangement and manufacturing sequence were directly motivated.
- Specific technical elements claimed as unique include the strap-mounted package and coil, coplanar/low-profile arrangement, and the particular hermetic package construction with mixed flip-chip and wire-bonded circuitry.
- Same-assignee sibling filings show a design-space exploration around retinal electrode arrays, side-mounted coils, implant packages, bonding, feedthroughs, and flexible circuits. This patent occupies the low-profile ocular integration / hermetic-package region of that design space.

Unexpected-result gate

No comparative performance result, survival data, hermeticity test data, or clinical outcome is established by the supplied record. The proposition enters the work layer with barrier type `insufficient_technical_demonstration`.

Regulatory scope estimate

The technology is an implantable medical device. FDA's Overview of Device Regulation identifies device classification, establishment registration/listing, premarket pathways, IDE requirements for clinical studies, QMSR/21 CFR Part 820, labeling, and medical-device reporting as applicable regulatory controls. Scope-level burden is **High**, but product classification and pathway selection require device-specific regulatory analysis.

Development stage

Patent-disclosed device architecture. Engineering feasibility is plausible from the described manufacturing sequence, but prototype, validation, clinical, and production evidence are not established from this intake. Next milestones: hermeticity testing; fatigue and ocular-motion testing; biocompatibility; electrical stimulation bench tests; animal/clinical protocol; and regulatory pathway confirmation.

Classification candidates

Candidates, not established classifications for the complete invention:

- A61N 1/0543 — retinal electrodes (USPTO CPC scheme; candidate).
- A61N 1/36046 — stimulation of the eye (USPTO CPC scheme; candidate).
- A61N 1/37223 — circuits for electromagnetic coupling (USPTO CPC scheme; candidate).
- A61N 1/375 — constructional arrangements/casings (USPTO CPC scheme; candidate).
- A61N 1/3754 — feedthroughs (USPTO CPC scheme; candidate).
- A61N 1/3758 — packaging of components within the casing (USPTO CPC scheme; candidate).
- A61N 1/3787 — electrical supply from an external energy source (USPTO CPC scheme; candidate).

What remains unclear

1. Whether the package/coil arrangement was commercialized or clinically deployed. 2. Actual hermeticity, fatigue, corrosion, thermal, and stimulation performance. 3. Current ownership and enforceability beyond the public assignment/status records. 4. Any pre-filing public disclosure, offer for sale, or clinical disclosure.

Patent Landscape Analysis

Patent Landscape — US 8,527,057 B2

Search scope and sources

Primary source: Google Patents family and classification record for US8527057B2.

Secondary source: USPTO CPC scheme for A61N. Search window: foundational retinal prosthesis and implant-package art through 2005 priority, plus sibling and later family records through 2026 for design-space context.

Established landscape findings

Design-space position

The invention is a convergence point between three technical streams: (1) retinal electrode fixation and flexible arrays, (2) inductive power/data coupling, and (3) hermetic miniaturized implant electronics. The same-assignee references indicate progressive exploration of individual modules before integration into the strap-mounted low-profile ocular configuration. This is contextual evidence, not a patentability score.

Quantitative landscape

The Google Patents XHR query `country=US&q=A61N1/0543 retinal` returned 492 US publication records. Returned applicant counts were Second Sight Medical Products 160, Nidek 42, Pixium Vision 32, Cochlear 28, and University of Southern California 25. The same query by jurisdiction returned US 492, WO 318, EP 261, CN 127, AU 105, JP 100, and CA 58 records. Frequency buckets returned by the query were 2016–2019: 64, 2019–2022: 43, 2022–2025: 38, and 2025–2028: 5.

These are directional publication-record counts from one public source, not deduplicated patent-family counts or normalized assignee groups. A second-source cross-validation remains a separate work item.

IP / Novelty Analysis

Novelty Search — US 8,527,057 B2

Claim-construction layer

The supplied patent provides formal claims. The principal evaluation target is claim 1, with a secondary check of the hermetic-package manufacturing disclosure. This is a preliminary analytical mapping, not a legal claim construction or FTO opinion.

- L1: electrode array suitable for mounting near a retina.
- L2: strap suitable for surrounding the sclera.
- L3: hermetic electronics package encasing a flip-chip circuit mounted to the strap.
- L4: electrical cable coupling electrode array and electronics package.
- L5: secondary inductive coil mounted to the strap, coplanar with the package, electrically coupled to it, powering the flip-chip circuit, and suitable for lateral scleral placement.

Flagged references

Claim-element mapping

No single external reference in the reviewed record discloses all five limitations. The anticipation proposition is therefore excluded from factual findings because a complete independent full-text/classification search was not completed in this run.

Causal Bridge Test — closest external pathway

```
`yaml causal_bridge_test: prior_state: known_objective: retinal electrical stimulation with implantable electrodes known_components: retinal arrays, inductive coils, hermetic implant packages known_principles: epiretinal stimulation, inductive power/data transfer, hermetic sealing claimed_state: claimed_configuration: low-profile scleral strap integrating a hermetic flip-chip package and coplanar receiving coil claimed_effect: compact implant with reduced height and protected electronics bridge: required_change: integrate the package, coil, array cable, fixation strap, and manufacturing stack into one ocular geometry bridge_evidence: direct_pre_filing: false analogous_pre_filing: true inventor_self_art: true general_science: true post_filing: excluded motivation: direct: [] analogous: [US 7,228,181;
```

US 2003/0109903; US 5,935,155] inferred: [reduce implant height and protect electronics]
contradicted: [] status: PARTIALLY GROUNDED expectation_of_success: technical:
medium unexpected_result: identified: false detail: comparative package, fatigue,
hermeticity, or clinical data not established bridge_status: TRAVERSED bridge_work_state:
EXHAUSTED `

Per-gate assessment

- Utility: supported at the architectural level by the detailed patent disclosure; not a clinical efficacy conclusion.
- Novelty: not established as a final conclusion because independent full-text search and continuity analysis remain incomplete.
- Inventive step: the specific integration step appears technically meaningful, but the evidence supports only a partially grounded bridge and does not establish an ultimate legal obviousness conclusion.

Literature Analysis

Literature Search — US 8,527,057 B2

Search scope

Queries covered retinal prosthesis, epiretinal implant, retinal electrode array, inductive retinal implant, hermetic implant package, and retinal-tack fixation. Primary literature source: PubMed. Adjacent technical/regulatory sources: NEI and FDA.

Literature disclosures

1. Margalit E, Maia M, Weiland JD, et al. "Retinal prosthesis for the blind." *Survey of Ophthalmology*. 2002;47(4):335–356. DOI: 10.1016/S0039-6257(02)00311-9. PubMed PMID 12161210. Review of neuronal electrical stimulation and retinal prosthesis architectures. Supports the background objective and epiretinal context; does not establish the target package geometry. 2. Luo YH, da Cruz L. "The Argus II Retinal Prosthesis System." *Progress in Retinal and Eye Research*. 2016;50:89–107. DOI: 10.1016/j.preteyeres.2015.09.003. PubMed PMID 26404104. Reviews the Argus II system and its clinical/regulatory significance. Supports the existence of a commercial retinal-prosthesis pathway; does not establish embodiment of every claimed package limitation. 3. Goetz GA, Palanker DV. "Electronic approaches to restoration of sight." *Reports on Progress in Physics*. 2016;79(9):096701. DOI: 10.1088/0034-4885/79/9/096701. PubMed PMID 27502748. Reviews epiretinal and subretinal approaches. Supports field orientation and technology alternatives. 4. Ghani N, Bansal J, Naidu A, Chaudhary KM. "Long term positional stability of the Argus II retinal prosthesis epiretinal implant." *BMC Ophthalmology*. 2023;23:70. DOI: 10.1186/s12886-022-02736-w. PubMed PMID 36797684. Addresses long-term implant positional stability and is relevant to fixation/implantation risk; it does not prove the specific US8527057 package was used. 5. Palanker D. "Electronic Retinal Prostheses." *Cold Spring Harbor Perspectives in Medicine*. 2023;13(8):a041525. DOI: 10.1101/cshperspect.a041525. PubMed PMID 36781222. Review of retinal prosthesis architectures and stimulation targets; useful background, not a claim-level disclosure of the target package.

Literature conclusion

The literature confirms the technical objective, retinal stimulation context, and clinical importance of fixation and implant stability. It does not, from the reviewed records, establish the complete strap + coplanar coil + hermetic flip-chip package combination. A broader engineering-database search for hermetic implant packaging and manufacturing disclosures remains a work item.

Market Analysis

Market Analysis — US 8,527,057 B2

Market boundary

The direct opportunity is implantable retinal-prosthesis systems for severe retinal degeneration, including the implant, external processor, surgical tools, clinical program, and service/support layer. Adjacent markets include implantable neural stimulation packaging, ophthalmic surgical implants, and other hermetic microelectronic medical devices.

Established market-relevance findings

- NEI describes retinitis pigmentosa as a group of rare retinal diseases that cause progressive vision loss and states that there is no cure, while vision aids and rehabilitation can help. Source: <https://www.nei.nih.gov/learn-about-eye-health/eye-conditions-and-diseases/retinitis-pigmentosa>
- PubMed literature identifies Argus II as a retinal-prosthesis system that entered clinical/commercial context and reviews epiretinal implant stability. Sources: PMID 26404104 and PMID 36797684.
- FDA states that medical-device manufacturers face classification, registration and listing, premarket, IDE, QMSR, labeling, and MDR controls. Source: <https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/overview-device-regulation>

Quantitative context

The verified Google Patents retinal-electrode query returned 492 US publication records, with 160 returned under Second Sight Medical Products, 42 under Nidek, 32 under Pixium Vision, 28 under Cochlear, and 25 under USC. These are directional publication-record counts, not deduplicated families or normalized corporate groups.

The verified adjacent manufacturing context is 2022 Census CBP NAICS 334510: 923 establishments, 92,285 employees, and annual payroll of \$10,030,531 thousand. The CBP file does not establish retinal-prosthesis revenue, shipment value, reimbursement, or market size. Employee-size bracket counts are included in the report chart and ledger. The

Census API key requirement prevented use of the API endpoint, so the downloadable CBP file was used instead.

Opportunity structure

- Product: implantable retinal-prosthesis subsystem or licensed packaging architecture.
- Buyer: medical-device company, clinical-development sponsor, or specialized implant manufacturer; end user is a patient treated through an ophthalmic surgical program.
- Channel: regulated medical-device development, clinical centers, and specialist ophthalmology distribution.
- Accessibility: low to medium because the technology requires clinical validation, manufacturing controls, regulatory submissions, and specialized implantation.
- Competitive intensity: high within retinal prostheses and implantable neurotechnology, with alternative epiretinal, subretinal, photovoltaic, and cortical approaches.

SWOT

Strengths: compact integrated architecture; detailed hermetic-package disclosure; strong fit with an identified retinal-prosthesis clinical need.

Weaknesses: no performance, clinical, production, or current ownership evidence in the supplied record; high manufacturing and regulatory complexity.

Opportunities: licensing package/process know-how to retinal-prosthesis or implantable neural-device developers; adapting package techniques to other ocular implants.

Threats: competing retinal-prosthesis architectures; clinical failure or tissue reaction; long development cycles; current legal-status and ownership uncertainty.

Commercial actionability

Commercial actionability is **Indeterminate-to-Limited** on the current evidence. The technical need is established, but commercialization readiness and quantitative market size are not established. This is not a financial forecast or legal opinion.

Potential Partners

Partner Identification — US 8,527,057 B2

Candidate pathways

No exclusivity claim is made. These are the strongest identified pathways from the available public record, not the only possible partners. Contact information and current interest require independent verification. Personal decision-maker profiles, LinkedIn identities, and a complete multi-company contact list were not established in this run; the report now includes only public contacts that were directly verified.

Operational Audit

Avenue Ledger — US8527057 v1.6 Run

Run date: 2026-08-15 **Submission:** submission-us8527057.md **Rule:** EXHAUSTED
is a work-layer termination state, not evidence.

Landscape & Market Data classification gate

The renderer must not substitute B25D, NAICS 332216, or any other unrelated class.

Evidence Recovery Record

v1.7 Evidence Recovery Record — US8527057

Patent-family archaeology

- US7881799B2 is the parent record for the target divisional.
- Google Patents lists the parent as active with adjusted expiration 2028-03-14.
- The family record shows priority to US8527057B2 and assignment events from Second Sight Medical Products to Vivani Medical in 2022 and Vivani to Cortigent in 2023.
- Target US8527057B2 is recorded separately as expired for maintenance-fee nonpayment.

Sources:

- <https://patents.google.com/patent/US7881799B2/en>
- <https://patents.google.com/patent/US8527057B2/en>

Close pre-critical-date references

- Priority: 2004-04-06, before the target's 2005-04-28 priority date.
- Subject matter: retinal electrode array, side-mounted secondary inductive coil, scleral strap, lateral placement, external inductive coupling, surgical handling features, and flexible cable arrangements.
- Status: close technical-lineage reference; no complete claim-1 anticipation finding is admitted because the reviewed record does not establish the target's full hermetic flip-chip package and all claimed relationships in one qualifying reference.

Source: <https://patents.google.com/patent/US20050222624A1/en>

- Priority: 2002-01-17.
- Subject matter: minimally invasive retinal prosthesis, external electronics/coils, flexible ocular geometry, and surgical placement constraints.
- Role: domain-level coverage and technical context; not a complete anticipation mapping.

Source: <https://patents.google.com/patent/US20030158588A1/en>

- Priority: 2001-11-16.

- Subject matter: flexible/stretchable retinal electrode arrays, conductive leads, polymer substrates, embedded electrodes, manufacturing, and flip-chip attachment.
- Role: product/process-domain coverage; not a complete anticipation mapping.

Source: <https://patents.google.com/patent/US20030097166A1/en>

Product and regulatory breadcrumb

FDA HDE H110002 identifies:

- Trade name: ARGUS II RETINAL PROSTHESIS SYSTEM.
- Applicant: Cortigent, Inc.
- Decision date: 2013-02-13.
- Product code: NBF, prosthesis retinal.
- Clinical trial: NCT00407602.
- Indication: adults age 25 or older with severe-to-profound retinitis pigmentosa, subject to light-perception, prior useful-vision, lens-status, follow-up, fitting, and rehabilitation criteria.

Source: <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfHDE/hde.cfm?id=H110002>

Clinical history breadcrumb

ClinicalTrials.gov search results for “Argus II” identify:

Source: <https://clinicaltrials.gov/api/v2/studies?query.term=Argus%20II&pageSize=20>

Remaining recovery work

- Map US8527057 claim limitations against the Argus II product and FDA device records.
- Retrieve examiner/applicant citations and prosecution history for the parent/divisional family.
- Reconstruct all continuation, divisional, foreign, and surviving family rights.
- Build a bounded patient/procedure/reimbursement model rather than a generic industry proxy alone.
- Determine whether commercialization failure, reimbursement, regulatory constraints, product economics, or business-model factors drove the post-market outcome.

Bounded patient-population model

PubMed review PMID 29597005 reports worldwide retinitis-pigmentosa prevalence of approximately 1:4,000. The model therefore begins with:

`text Population with RP $\approx$ reference population $\div$ 4,000 Potential severe/profound candidates = RP population $\times$ eligibility fraction Addressable implants = eligible candidates $\times$ diagnosis/access/consent fraction Economic opportunity = addressable implants $\times$ device/procedure economics `

The FDA HDE record supplies the eligibility constraints, but the eligible fraction, diagnosis/access fraction, device price, procedure reimbursement, and adoption rate are not established. The model is therefore a recovery scaffold and sensitivity framework, not a TAM/SAM/SOM claim.

Source: <https://pubmed.ncbi.nlm.nih.gov/29597005/>

Rights state

- Patent: US8527057B2
- State: **EXPIRED**
- Active: **False**
- Status source: <https://patents.google.com/patent/US8527057B2/en>
- Status disclaimer: public database status is an assumption and not a legal conclusion.
- Target-patent licensing: **constrained by target lapse**.
- Family-level licensing: **not blocked**; active family members shown in the current record include US7881799B2 (expires 2028-03-14) and US8473048B2 (expires 2028-06-25).
- Required strategy: family/portfolio and surviving-rights diligence before any licensing recommendation.

Anticipation state

UNRESOLVED — SEARCH-INCOMPLETE. No positive novelty conclusion is emitted. The remaining work is tracked under P-05-001 and P-05-002 with escalation-required state.

Bridge vector

PARTIALLY TRAVERSED — feature availability HIGH; motivation MODERATE; compatibility PARTIAL; architecture constraint STRONG; expected and unexpected result NOT ESTABLISHED.

Evidence debt and downstream constraints

The compiled graph contains 4 evidence-debt items and 5 downstream constraints. The v1.7 renderer must display these constraints beside any derived score or recommendation.

Evidence recovery queue

The controller has converted the v1.6 stopping points into explicit recovery work:

Claim-domain vectors

The claim graph separates retinal interface, mechanical fixation, electronics packaging, interconnect, power architecture, manufacturing process, and surgical-handling domains. Product and process vectors are evaluated independently; the cross-domain spatial integration is retained as an inventive-center candidate, not promoted to a finding.

Technology lifecycle queue

Argus II and successor-company breadcrumbs now trigger a required lifecycle review: prototype → clinical validation → regulatory pathway → commercialization → market outcome → failure mode → technology-resurrection opportunity. The current v1.7 run does not classify the failure mode until those recovery paths are executed.

Recovery evidence obtained in this run

- **Family archaeology:** US7881799B2 is the parent record and is listed by Google Patents as active with an adjusted expiration of 2028-03-14; the same record shows priority to US8527057B2 and assignments from Second Sight to Vivani in 2022 and Vivani to Cortigent in 2023.
- **Close pre-critical-date reference:** US20050222624A1, *Retinal prosthesis with side mounted inductive coil*, has a 2004-04-06 priority date and teaches an electrode array, side-mounted secondary coil, scleral strap, lateral placement, and external inductive coupling. It is a close technical-lineage reference, not by itself a complete anticipation mapping for US8527057 claim 1's hermetic flip-chip package.
- **Additional pre-critical-date references:** US20030158588A1 covers a minimally invasive retinal prosthesis; US20030097166A1 covers flexible retinal electrode arrays and manufacturing. These support domain-level coverage but do not establish all claim-1 limitations in one reference.
- **Product/regulatory breadcrumb:** FDA HDE H110002 identifies the ARGUS II RETINAL PROSTHESIS SYSTEM, applicant Cortigent, decision date 2013-02-13, product code NBF, clinical trial NCT00407602, and an indication for adults with severe-to-profound retinitis pigmentosa meeting specified eligibility criteria.
- **Clinical breadcrumb:** ClinicalTrials.gov search results identify Argus II studies including NCT00407602 (completed feasibility), NCT01860092 (terminated post-approval enrollment), NCT01490827 (terminated post-market surveillance), NCT02303288 (completed France post-market study), and NCT04359108 (completed environmental localization study). These establish product/clinical history, not embodiment of every US8527057 limitation.

These resolved breadcrumbs reduce the commercialization/product-identity unknown, but claim-to-product mapping, prosecution-history review, family-wide surviving-rights analysis, reimbursement economics, and failure-mode classification remain explicitly queued rather than silently upgraded.

Bounded patient model

PubMed PMID 29597005 reports worldwide retinitis-pigmentosa prevalence of approximately 1:4,000. The controller has therefore created a bounded model rather than a fabricated market size: reference population $\div$ 4,000 $\rightarrow$ severe/profound eligibility fraction $\rightarrow$ diagnosis/access/consent fraction $\rightarrow$ device/procedure economics. The FDA HDE supplies eligibility constraints, but the fractions, pricing, reimbursement, and adoption inputs remain unresolved and continue to cap commercial confidence.